

# DSAA3073: Theories in Data Science

## Week 5 Validation Sheet · Stability, Tikhonov, and Ridge

**Coverage:** Week 5

**Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

### Part A: Multiple choice

**A1 [5 marks].** At a fixed input  $x_0$ , which condition is needed to remove a cross term between the fitted value  $\hat{f}_{S(x_0)}$  and fresh test noise  $\varepsilon$  in the squared-loss bias-variance decomposition?

- (a)  $\mathbb{E}[\varepsilon] = 0$  is always sufficient.
- (b)  $\mathbb{E}[\varepsilon \mid S, x_0] = 0$ , as supplied by fresh zero-mean noise.
- (c) The training predictions have zero variance.
- (d) The fitted value equals  $f^*(x_0)$  for every sample.

**A2 [5 marks].** Compare  $A_0 = \text{diag}(1, 0)$  and  $A_\varepsilon = \text{diag}(1, \varepsilon)$  with  $\varepsilon > 0$ . Which statement is correct on  $\mathbb{R}^2$ ?

- (a) Both have a discontinuous inverse.
- (b)  $A_0$  has a unique inverse, while  $A_\varepsilon$  may have none.
- (c)  $A_0$  can fail existence or uniqueness;  $A_\varepsilon$  is invertible but has inverse gain  $\frac{1}{\varepsilon}$  in its second direction.
- (d) A small nonzero singular value is mathematically identical to a zero one.

**A3 [5 marks].** Why does the finite-dimensional objective  $J_{\lambda(f)} = \|Af - y\|_2^2 + \lambda\|f\|_2^2$  have one global minimizer for every matrix  $A$  when  $\lambda > 0$ ?

- (a) Every convex problem has a unique minimizer.
- (b)  $A^T A + \lambda I$  is positive definite.
- (c) The residual can always be made zero.
- (d) The penalty removes the noise from  $y$ .

**A4 [5 marks].** In a positive eigendirection of  $X^T X$  with eigenvalue  $\mu$ , what fraction of the OLS coefficient is retained by ridge at  $\lambda > 0$ ?

- (a)  $\frac{1}{\mu + \lambda}$
- (b)  $\frac{\lambda}{\mu + \lambda}$
- (c)  $\frac{\mu}{\mu + \lambda}$
- (d) Either 0 or 1, because ridge performs subset selection.

## Part B: Diagnose a repeated-fit system

**B1 [20 marks].** Three systems have the same training MSE. At one test input,  $f^{\star}(x_0) = 4$ . The table reports four independent refit predictions, the eigenvalues of the normalized Gram matrix, the fraction of refit movement aligned with the weakest eigendirection, and the fresh-noise standard deviation.

| system | refits             | spectrum       | weak share | $\sigma$ |
|--------|--------------------|----------------|------------|----------|
| P      | 3.7, 4.1, 3.9, 4.3 | (1, 0.7, 0.4)  | 20%        | 0.2      |
| Q      | 2.0, 5.5, 2.5, 6.0 | (1, 0.5, 0.01) | 90%        | 0.2      |
| R      | 2.8, 3.2, 3.0, 3.0 | (1, 0.8, 0.6)  | 15%        | 0.2      |

Diagnose the dominant problem in each system. For each one, defend or reject ridge as the first intervention using at least two columns of evidence. State the most that the Week 5 evidence can justify without choosing a numerical value of  $\lambda$ .

*Answer space*

## Part C: Basic calculations

**C1 [7 marks].** At fixed  $x_0$ , four equally likely training samples produce predictions 2, 4, 4, 6. If  $f^{\star}(x_0) = 5$  and the fresh-noise variance is 3, compute the squared bias, training-sample variance, and expected squared prediction error. Use population averaging over the four declared outcomes.

*Answer space*

**C2 [8 marks].** In one positive eigendirection of  $X^T X$ , let  $\mu = 4$ ,  $\lambda = 1$ , the true coefficient be  $\theta = 2$ , and the noise variance be  $\sigma^2 = 3$ . Compute the ridge shrinkage factor, the mean ridge coefficient, its bias, and its variance.

*Answer space*

## Part D: Construct and verify boundary cases

**D1 [17 marks].** Consider  $A_0 = \text{diag}(2, 0)$  and  $A_s = \text{diag}(2, 0.02)$  on  $\mathbb{R}^2$ .

For  $A_0$ , construct one observation with no exact solution and a different observation with infinitely many exact solutions, verifying both claims. For  $A_s$ , construct a perturbation of norm 0.002 whose recovered change has norm 0.1. Use the constructions to explain why a zero singular value and a small positive singular value are not the same failure.

*Answer space*

**D2 [18 marks].** Let

$$A = \text{diag}\left(2, \frac{1}{2}\right), \quad y = (4, 1), \quad \lambda = 1,$$

and  $J(f) = \|Af - y\|_2^2 + \|f\|_2^2$ . Derive the unique minimizer over all  $f \in \mathbb{R}^2$ . Then verify, rather than merely assert, that it is the global minimizer. Finally compare its two coordinates with those of the exact inverse  $A^{-1}y$ .

*Answer space*

## Part E: Transfer to a coupled inverse

E1 [25 marks]. Let

$$v_+ = \frac{1,1}{\sqrt{2}}, \quad v_- = \frac{1,-1}{\sqrt{2}}, \quad u_1 = (1,0), \quad u_2 = (0,1).$$

These are orthonormal bases of the input and observation spaces. For  $\tau \in \{0,0.1\}$ , a linear map is defined by

$$A_\tau v_+ = 2u_1, \quad A_\tau v_- = \tau u_2.$$

The observed vector is  $y = 2u_1 + 0.2u_2$ . For  $\lambda > 0$ , define

$$J_{\tau,\lambda}(f) = \|A_\tau f - y\|_2^2 + \lambda \|f\|_2^2.$$

Determine the unique minimizer  $f_{\tau,\lambda}$ , expressed in standard coordinates, for each value of  $\tau$ . Use your result to compare the two maps' treatment of the  $u_2$  observation component and to determine both limits as  $\lambda \rightarrow 0$ . Explain precisely which limit is an exact solution and which is the Moore–Penrose minimum-norm least-squares solution.

*Answer space*

## Part F: Limit a claim before selecting a parameter

**F1 [15 marks].** An analyst increases  $\lambda$  and observes two facts: the minimized training residual becomes larger, and the spread of predictions across repeated fits becomes smaller. They conclude, “test error must therefore have decreased, so these observations are enough to choose this value of  $\lambda$ .”

Audit the conclusion using only Week 5 concepts. State what the two facts do establish, identify the missing quantities, and explain why Week 5 diagnosis alone cannot honestly select a numerical  $\lambda$ .

*Answer space*